

COURSE SYLLABUS

CSC14006 – Pattern Recognition

1. GENERAL INFORMATION

Course name:	Pattern Recognition
Course name (in Vietnamese):	Nhận dạng
Course ID:	CSC14006
Knowledge block:	Specialize
Number of credits:	4
Credit hours for theory:	45
Credit hours for practice:	30
Credit hours for self-study:	90
Prerequisite:	
Prior-course:	Introduction to artificial intelligence
Instructors:	Lê Hoàng Thái

2. COURSE DESCRIPTION

The course is designed to provide students:

1. The foundations of pattern recognition, i.e. how patterns are recognized on the computer.
2. Wide understanding of different related topics, i.e., pattern recognition systems, preprocessing and feature extraction, theories of supervised and unsupervised learning, object classification and recognition, deep neural networks, and other topics as well.
3. With sufficient background in the theoretical concepts behind stages of pattern recognition systems, and to implement these systems using computer applications.

3. COURSE GOALS

At the end of the course, students are able to

ID	Description	Program LOs
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G1	Describe about pattern recognition and its broad applications in various aspects of our day-to-day life.	1.4.1, 3.1.1
G2	Explain the algorithms used in different phases of pattern recognition systems, including data acquisition, pre-processing, segmentation, feature extraction, and classification.	1.4.1, 4.3.2, 4.3.3, 4.3.4
G3	Demonstrate the techniques used to recognize patterns, such as statistical approaches, data clustering, neural networks, etc.	1.4.1
G4	Summarize knowledge of various applications of pattern recognition in real life, including reading research papers and preparing presentations.	1.4.1, 2.3.2, 2.4.5, 3.1.1, 4.2.2
G5	Produce pattern recognition programs and implement various techniques discussed throughout the course.	1.2.1, 1.4.1, 5.3.2

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1	Summarize the concepts of pattern recognition, including its definition, design cycle, applications, and algorithms.	T
G1.2	Explain basic concepts involved in structural and statistical pattern recognition.	T
G2.1	Experiment pattern recognition systems.	U
G2.2	Compare and analyze algorithms used in pattern recognition.	U
G2.3	Explain various techniques used in object detection and classification.	T
G3.1	Explain various techniques used in pattern recognition systems, including supervised and unsupervised learning.	T

G3.2	Apply mathematical tools to algorithm design.	U
G4.1	Translate abstract ideas into practice.	T, U
G4.2	Show the ability to read research papers.	U
G4.3	Show the ability for oral presentations.	U
G5.1	Describe the important principles of deep neural networks, support vector machine, Principal Component Analysis, Linear Discriminant Analysis, and local binary pattern.	T
G5.2	Show a good programming style.	U

5. TEACHING PLAN

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)
1	Class overview: Class organization, topics overview, software etc. Pattern recognition: Introduction to Biometric Systems	G1.1, G1.2, G4.1	Lecturing Q&A, Group discussion QZ1: Quiz 1 (Chapter 1)
2	Various phases of pattern recognition systems	G2.1, G2.2, G2.3, G4.1	Lecturing Demonstration, Q&A QZ2: Quiz 2
3	Feature extraction, object detection and classification using Local binary pattern (LBP)	G2.2, G2.3, G3.1, G3.2	Lecturing Demonstration, discussion
4	Feature extraction using Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA).	G2.2, G2.3, G3.1, G3.2	Lecturing Demonstration, discussion

5	Data clustering, Classification and Template matching (algorithms: K-mean, k-nearest neighbor and Support vector machine).	G2.2, G2.3, G3.1, G3.2	Lecturing Demonstration
6	Various phases of Face Recognition System, including face detection, face alignment, feature extraction, feature matching and their applications.	G2.1, G2.2, G2.3, G3.1, G3.2, G4.1	Question & answer Case study and discussion QZ3: Quiz 3
7	An introduction to Deep Learning: Deep learning architectures	G1.1, G1.2, G2.1, G2.2, G2.3, G3.1, G3.2, G4.1	Lecturing Demonstration
8	Deep Learning Tutorial: Why Deep? Tips for Training Deep Neural Network.	G1.1, G1.2, G2.1, G2.2, G2.3, G3.1, G3.2, G4.1	Lecturing Q&A, discussion QZ4: Quiz 4
9	Deep learning architectures: convolutional neural networks (CNNs) and their applications.	G1.1, G1.2, G2.1, G2.2, G2.3, G3.1, G3.2, G4.1	Lecturing
10	Presentations	G4.1, G4.2, G4.3, G5.1, G5.2	Case study, discussion Demonstration
11	Review	G5.2	Lecturing Q&A, Discussion Project submitted

For the practical laboratory work, there are 10 weeks which cover similar topics as it goes in the theory class. Each week, teaching assistants will explain and demonstrate key ideas on the

corresponding topic and ask students to do their lab exercises either on computer in the lab or at home. All the lab work submitted will be graded. There would be a final exam for lab work.

6. ASSESSMENTS

ID	Topic	Description	Course outcomes	Ratio (%)
A1	Assignments			20%
A11	Quizzes: QZ1, QZ2, QZ3, and QZ4.	Small quizzes in class for each topic	G1.1, G1.2, G2.1, G2.2, G2.3, G3.1, G3.2, G4.1	5%
A12	Homework: HW1, HW2, and HW3	HW1, HW3: reading comprehension and writing reports in English HW2, HW3: practicing based on knowledge taught in class	G4.1, G4.2, G4.3, G5.1, G5.2	5%
A13	Weekly labwork: LW1–LW10		G4.1, G4.2, G4.3, G5.1, G5.2	10%
A2	Projects			20%
A21	Project		G4.1, G4.2 G4.3, G5.1, G5.2	20%
A3	Exams			60%
A31	Lab final exam	In-class programming exam on computer	G5.2	10%
A32	Presentations	Oral presentations with slides.	G4.1, G4.2, G4.3, G5.1, G5.2	20%

		Describe the understanding of different topics.		
A33	Final exam	Closed book exam. Describe the understanding of different topics, analyze & program to solve problems	G5.1, G5.2	30%

7. RESOURCES

Textbooks

- *Pattern Classification 2nd Edition*, Richard O. Duda, Peter E. Hart, and David G. Stork Wiley, 2004.
- *Handbook of Biometrics*, Anil K. Jain, Patrick Flynn, and Arun A. Ross, Springer, 2008.
- *Handbook of Face recognition*, Stan Z. Li and Anil K. Jain, Springer, 2011.
- *Deep Learning for Computer Vision with Python, Starter Bundle*, Dr. Adrian Rosebrock, 2017.
- *Deep Learning for Computer Vision with Python, Practitioner Bundle*, Dr. Adrian Rosebrock, 2017.

Others

- <http://www.pyimagesearch.com/contact>

8. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students who are absent for more than 3 theory sessions are not allowed to take the exams.
- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.

- Students are encouraged to form study groups to discuss on the topics. However, individual work must be done and submitted on your own.

Ho Chi Minh City, February 13th, 2023

DEAN OF FACULTY

A handwritten signature in blue ink, appearing to read 'Dinh Ba Tien', with a horizontal line underneath.

Dinh Ba Tien